Assimilação de Dados por Redes Neurais Celulares

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Sumário

- Cellular Neural Networks (CelNN);
- Assimilação de dados por CelNN;
- Assimilação de dados no modelo de Lorenz-63;
- Assimilação de dados de radiância no modelo CAM-CESM2;
- Resultados;
- Perspectivas.

Cellular Neural Networks

As CelNNs são especialmente adequados para processar dados em domínios espaciais, como imagens, onde a conectividade local e o processamento simultâneo são essenciais.

Cellular Neural Networks

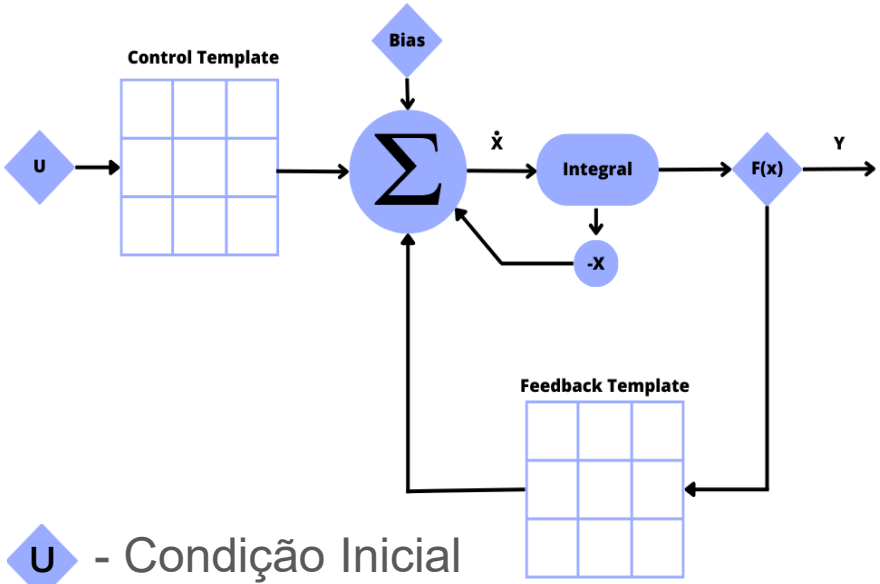
$$\dot{x} = -x + T_C u + T_f y + I$$

60	113	56	139	85
73	121	54	84	128
131	99	70	129	127
80	57	115	69	134
104	126	123	95	130

Template

0	-1	0
-1	5	-1
0	-1	0

114				

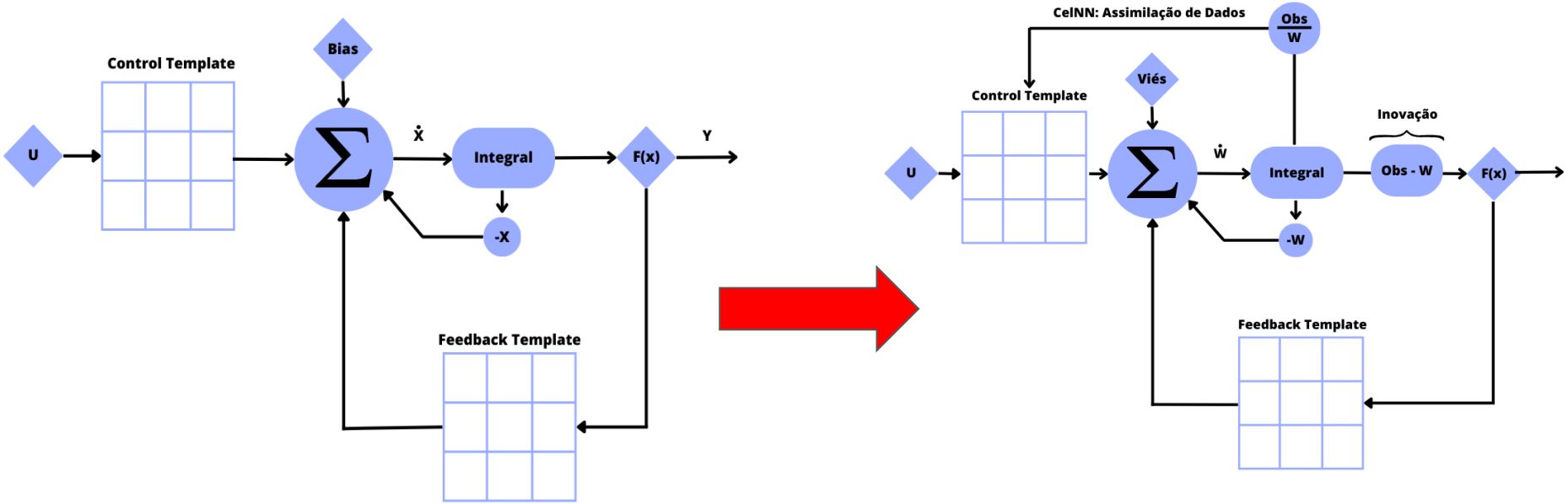


U - Condição Inicial

X - Estado Atual da Célula

Y - Saída

Assimilação de dados por CelNN



https://www.sciencedirect.com/science/article/abs/pii/S187775032500064X

Assimilação de dados por CelNN

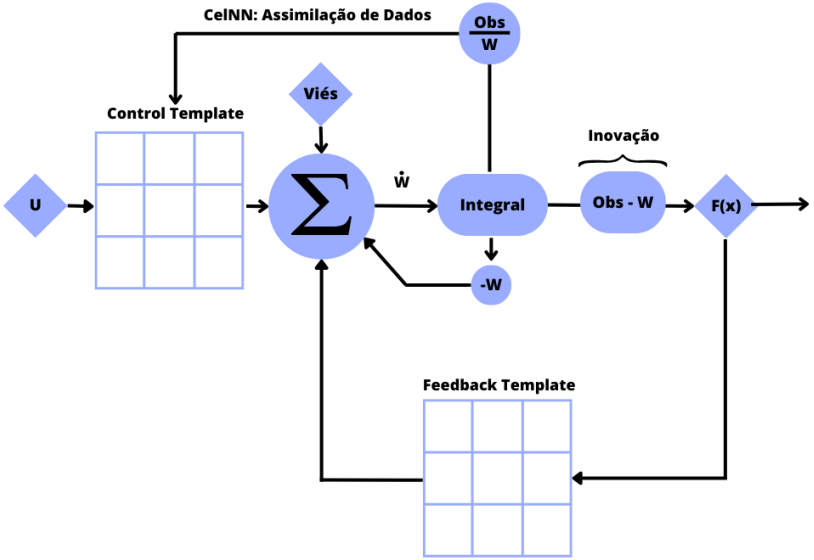
U =

15	12	10
9	19	7
6	10	8

Obs = 17

W_t =

16	11	9
10	18	8
6.5	9	7.5



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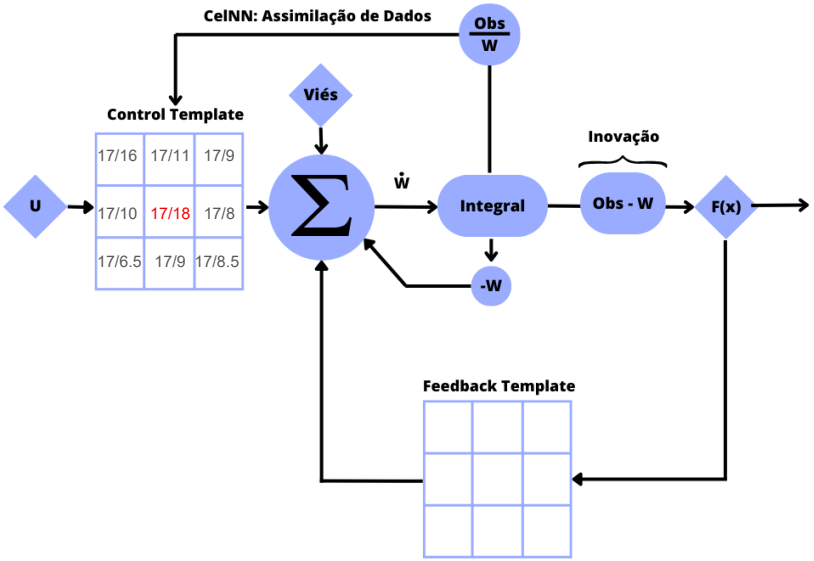

Assimilação de dados por CelNN

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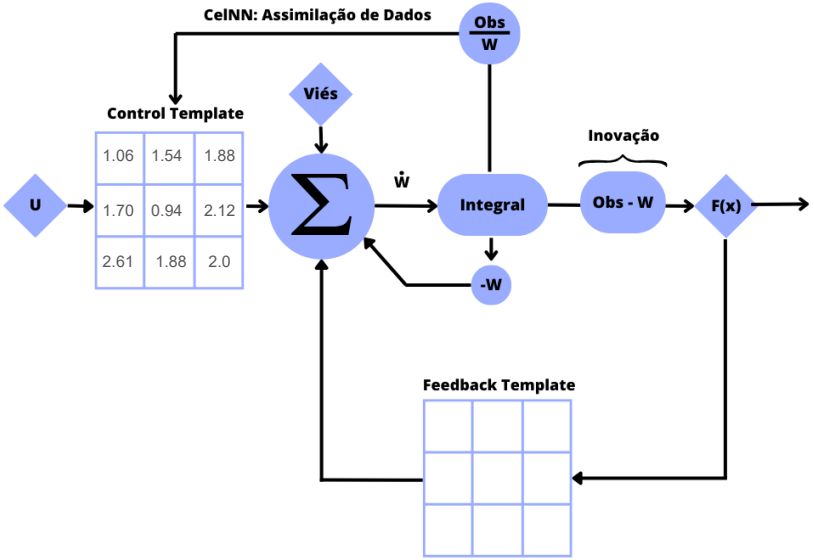

Assimilação de dados por CelNN

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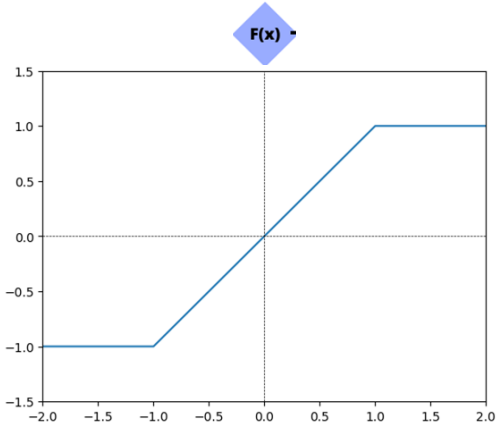
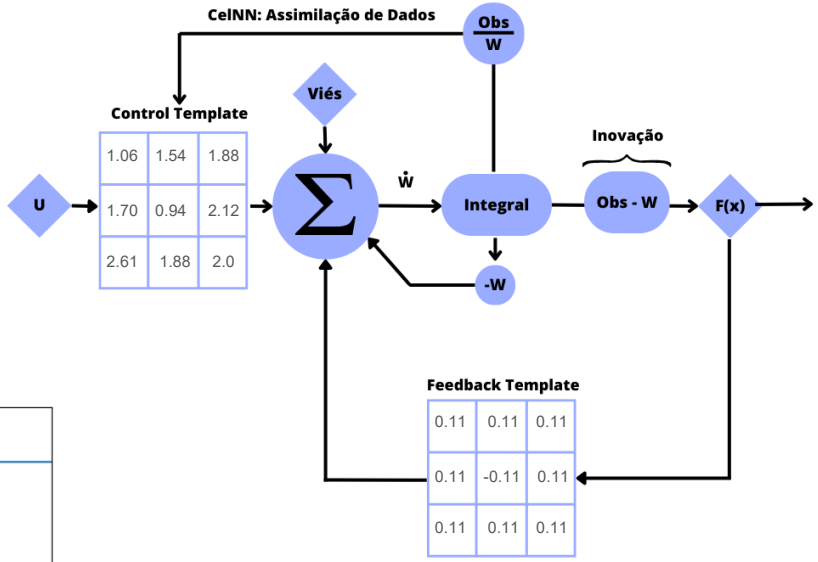




Assimilação de dados por CelNN

U =

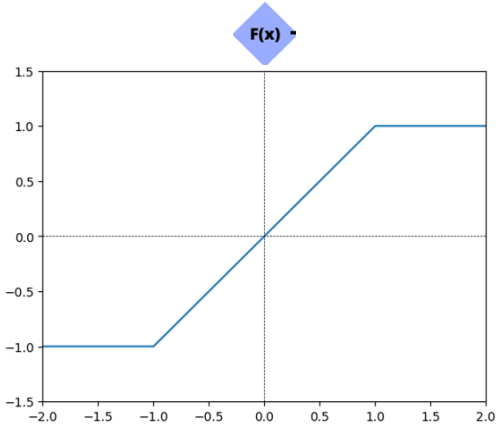
15	12	10
9	19	7
6	10	8

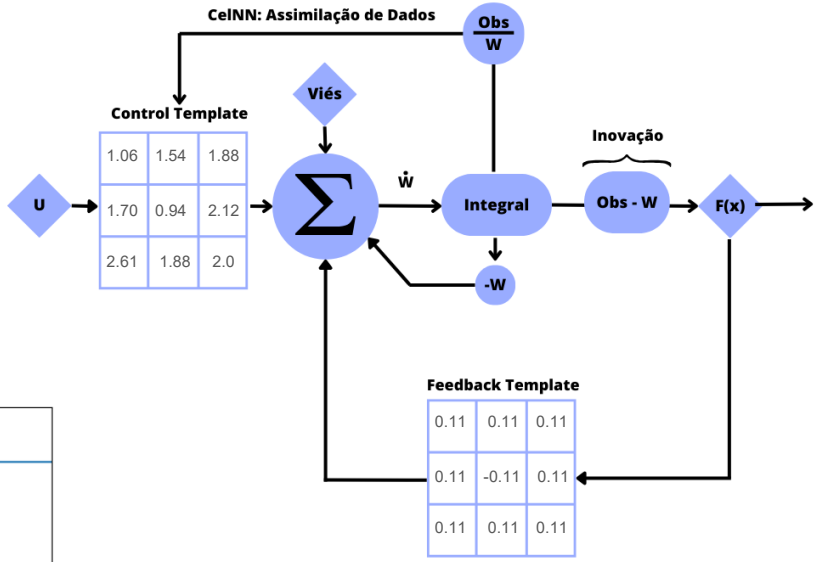
Obs = 17

W_t+1 = 17.63

W_t =

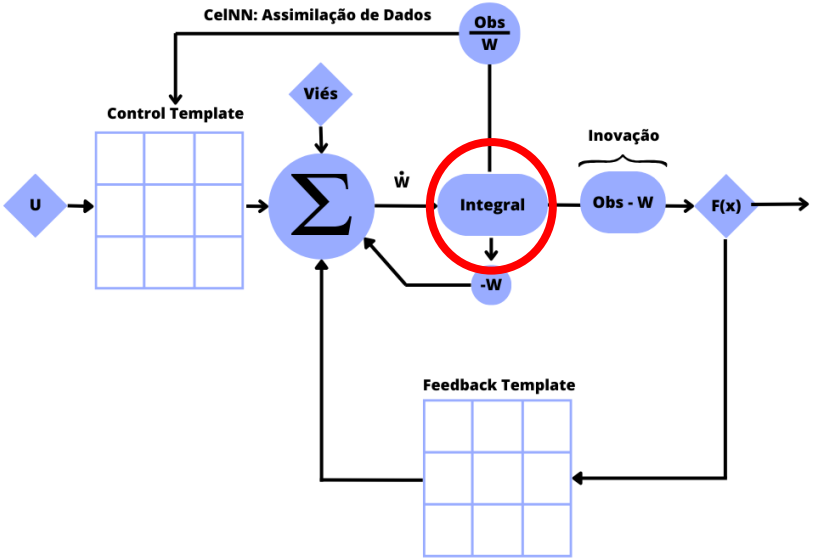
16	11	9
10	18	8
6.5	9	7.5





Cellular Neural Networks

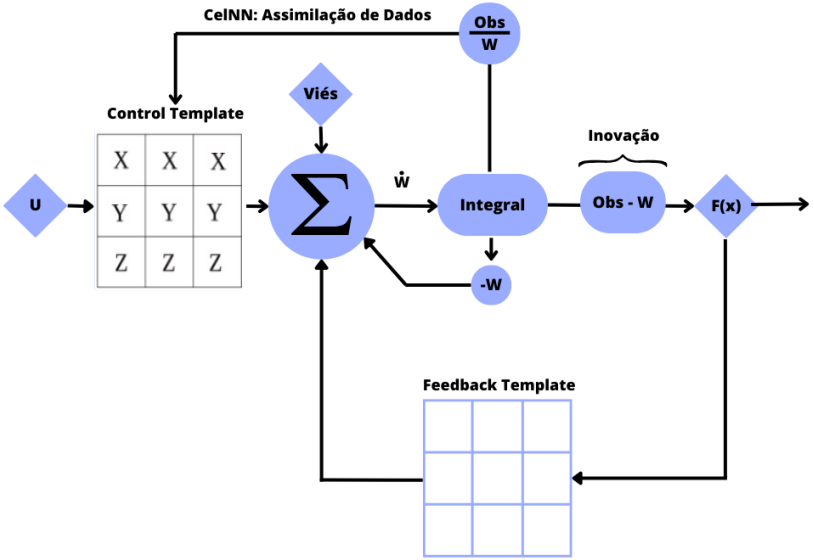
$$\dot{x} = \underbrace{-x}_{W_t = 18} + \underbrace{T_C u + T_f y}_{W_t+1 = 17.63} + I$$



Assimilação no modelo de Lorenz-63

$$\frac{dX}{dt} = \sigma(Y - X)$$
$$\frac{dY}{dt} = X(\rho - Z) - Y$$
$$\frac{dZ}{dt} = XY - \beta Z,$$

Lorenz-63



Agradecimentos

Contato

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